

Isolon: Keeping Isolation Claims Honest in a Bare-Metal Type-1 Hypervisor

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Abstract

Isolon is a clean-slate Type-1 hypervisor that boots as one UEFI binary on a Dell PowerEdge R640. On that server it reaches unmodified Linux 6.12 and, on the same boot, virtio-blk, virtio-net, dual-vCPU SMP, and four-VM probes, then serves host-NIC HTTP.

A machine-checked isolation theorem next to that binary is a labelling problem: the proved artifact and the mapping path the guests actually use need not be the same object. We propose a two-axis ladder—proof maturity (L0–L3) against artifact (ghost model, executable module, binary on the target)—and apply it to Isolon and to seven prior systems. Isolon’s L3 result is guest-exclusivity in a host-only Verus crate. Linux’s 512 MiB identity window is a 16-slot range registry with no ghost correspondence; the proved 4K table is the demand-fill and self-test path. Hypervisor non-access is an assumption. Virtio rings are guest frames the emulator touches. Nested VT-x completed a distro install-to-disk; that host is not the R640.

CCS Concepts: • Software and its engineering → Software verification; Virtual machines; • Security and privacy → Logic and verification.

Keywords: formal verification, Type-1 hypervisor, EPT, Verus, maturity ladder, experience

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1 Introduction

A hypervisor’s job is to make one computer look like several, without letting those several computers look at each other. On modern Intel servers the mechanism that actually enforces that is Extended Page Tables (EPT) [11]. If EPT is wrong, two guests share a frame, or a guest sees hypervisor memory, and no amount of management-API polish repairs it.

Commercial Type-1 stacks—ESXi, Xen [1], KVM [12]—are large, mature, and tested. They are not accompanied by a machine-checked isolation theorem for the frames they hand to guests. Research systems that *are* proved (seL4 [13, 14],

CertiKOS [8], Komodo [6]) target microkernels, certified kernels, or enclaves. They are not a single UEFI binary aimed at the PowerEdge fleet already in the rack.

Isolon is a clean-slate Type-1 hypervisor written in Rust. It compiles to one .efi and boots on a Dell PowerEdge R640: Intel Xeon Silver 4110 at 2.10 GHz (family 6, model 0x55, stepping 4), BIOS 2.2.11, boot map pages_{above_1MiB}=16,574,029 (≈63.2 GiB), iDRAC virtual floppy, COM2. Formal verification is an architectural constraint, not a post-hoc whitepaper. We picked Verus [16, 17] for SMT proofs of ghost state and Kani [4, 15] for bounded checking of unsafe. The interesting problem is not that a bijection between two maps can be proved. It is that a project shipping a hypervisor will be tempted to put that bijection in the abstract and then spend the rest of the paper taking it back. We treated that temptation as the thing to study.

The proved artifact is exclusivity among guests in a host-only ghost model. Live executable EPT is a bounded ownership registry. Device emulation, HTTP, the SPA, ISO handling, and guest firmware are outside the Proven Core. Claiming “the hypervisor is proved” would be proving the wrong thing [13]: a convenient proxy.

This paper makes four contributions:

1. **A two-axis labelling ladder** (Section 3.4): proof maturity L0–L3 against artifact (ghost model / executable module / binary on the target). An L3 claim about a ghost model and an L3 claim about executable code are not comparable. We apply the grid to Isolon and to seven systems from the literature (Table 7).
2. **Five lessons that generalized** while shipping the EFI (Section 7): pin the prover, do not host SMT on a UEFI target, do not poke third-party firmware control state, firmware TCP is not a management plane, and nested VT-x is not the production server. Table 8 archives the negative results behind those lessons.
3. **Guest-exclusivity in a ghost model** for 4K map/unmap, violation dispositions, and page transfer. The modelling that was not free is the emulate/reject/claim trichotomy and 2M/1G span encoding.
4. **A split evaluation on one R640:** Linux 6.12 to a shell, virtio-blk, virtio-net, dual-vCPU SMP, four-VM

probes, host-NIC HTTP, and El Torito CD EFI, with nested VT-x as a separate column.

Isolon is not seL4 for hypervisors and is not a verified KVM. It is a hypervisor we boot on iron, a ghost model we discharge, and a ladder that stops those two facts from being fused in a sentence.

2 Threat Model and Non-Goals

2.1 Why memory isolation is the headline

A Type-1 hypervisor is the TCB for every guest it runs. CPUID and MSR firewalls, interrupt injection, and the hypercall door all matter. None of them matter if EPT can map the same host frame to two guests without anyone noticing. We therefore name *exclusivity among guests* as the property we prove, not “the guest seemed isolated in tests,” and not “the hypervisor never touches a guest frame.”

2.2 In scope

We assume:

- A **malicious or compromised guest** (ring-0 Linux, firmware, or installer) that may issue any architected VM exit: EPT violation, CPUID, MSR, I/O, hypercall, HLT, exception.
- **Buggy device emulation** outside the Proven Core (virtio, IDE/ATAPI, UART). Bugs there can DoS a guest or confuse an installer; they must not become a second EPT mapping of a Proven Core frame.
- **Operator error** (wrong ISO, exhausted USB stick). Audit events record what happened; they do not enlarge the theorem.

Access versus mapping (virtio). Isolon ships virtio-blk and virtio-net. Virtqueue rings live in guest-allocated frames. Hypervisor emulation at VMX root *must* read and write those frames. That is, by construction, a guest-owned frame touched by the hypervisor. Exclusive ownership in this paper is a statement about *EPT mappings*: a frame in *owned* has exactly one guest as EPT owner, and the hypervisor does not install a second EPT entry for that frame in another guest’s EPT. Transient, emulation-mediated access from VMX root is not a second owner. If “belongs to neither the hypervisor” is read as “the hypervisor CPU never loads from that frame,” the property is false of the product we ship.

DMA and passthrough. The adversary in this section is a malicious ring-0 guest. Such a guest can program DMA if a device is passed through, or if an emulated device can be driven to DMA outside the guest’s frames. Isolon’s product path has **no device passthrough**. Guest DMA is mediated by emulation executing in the hypervisor’s address space. Exclusive ownership is *conditional on that*: it is not a theorem

against an IOMMU-less passthrough device. Unmodelled DMA is listed in Section 2.3. Closing DMA as a lemma needs an IOMMU story we do not have.

We do *not* assume a correct third-party guest firmware. Forcing a foreign firmware’s WaitForEvent internals to skip a check produced a NULL-event page fault and a dead loop. Product ISO boot now uses firmware we author. That firmware is **outside** the Proven Core.

2.3 Assumptions, not theorems

- The Intel SDM’s VMX/EPT hardware behaves as documented [11].
- The frozen Verus/SMT stack is trusted for the ghost proofs.
- Host physical memory given to the allocator is the memory the machine has.
- Hypervisor frames are those *absent* from the guest ownership registry. We do not prove that the EFI’s own page tables never map a guest-owned frame, and we do not prove that a bug cannot install a guest leaf pointing at hypervisor .text.
- No device passthrough; guest DMA is emulation-mediated, as above.
- Nested VT-x on a lab server is not a proof about a PowerEdge R640.
- Full x86 ISA / VMCS host-state correctness is inside the Proven Core *budget* and is not at proof-complete maturity beside the EPT crate.

3 Isolon Architecture

3.1 Single binary

Everything compiles to one UEFI PE/COFF image. Kernel, initrd, Web UI, and schemas are compressed sections, decompressed on first use. The size target is 15 MB with a 20 MB hard limit. There is no host Linux, no package manager, and no config tree the operator must get right before the first serial line.

3.2 Proven Core boundary

Only the modules in Table 1 are in scope for Verus specifications and progressive proofs. Everything else uses Rust’s type system, testing, and review. Promotion into the Proven Core requires a written architecture decision; the default answer is no. The hard cap is 15,000 lines including proof scaffolding.

Lived split, as of this submission: the machine-checked exclusivity theorems live in a host-only crate (ept_model, 2,366 lines) that is *not* linked into the .efi. The executable

Table 1. Proven Core budget (cap, not an L3 claim per row).

Module	Est. LOC	Why inside
VMX lifecycle	800	VMXON/VMXOFF leave CPU mode
VMCS management	1,500	Host-state corruption
EPT engine	2,000	Isolation mechanism
Frame allocator	1,500	Double-alloc / UAF
vCPU state	1,000	Save/restore leaks
Interrupt injection	800	Guest privilege
Hypercall door	500	Only intentional channel
MSR/CPUID/CR firewalls	1,200	Host feature subversion
Audit-log integrity	600	Tamper-evident trail
IPI confinement	500	Cross-VM IPI
Cap (code + scaffolding)	15,000	—

EPT ownership registry is 705 lines at spec-written maturity. Device emulation, HTTP, the SPA, ISO, guest firmware, iDRAC, and migration tooling are outside.

3.3 Hardware

Dell PowerEdge R640 / R650 / R660. Low-risk, self-sufficient surfaces (serial via iDRAC virtual console, SMBIOS/ACPI topology, onboard LOM) are in scope. OEM RAID health schemas are not a blocker for the theorems or for the single-host product loop.

Code in the Proven Core is required to document invariants at the function boundary, keep unsafe blocks minimal, and tag each of them for bounded checking. Mutable globals require a lock and a justification comment. We use explicit state machines rather than boolean flags. These rules are how a codebase stays L3-shaped when most modules are still L1: the later proof should not require a rewrite of ownership.

3.4 Maturity ladder as a community instrument

Proof maturity is one axis:

- **L0** documented invariants in comments.
- **L1** runtime `assert! / debug_assert!` plus Kani on unsafe.
- **L2** Verus specifications with ghost state and contracts.
- **L3** cargo `verus` verify succeeds for the claimed surface.

The second axis is *which artifact* the claim is about: a ghost model, an executable module, or a binary on the target. Table 2 places Isolon. An L3 claim about a ghost model and an L3 claim about executable code are not comparable; a one-axis label “L3 isolation” would make Isolon’s crate look like seL4’s C kernel. Section 6.5 applies the same grid to seven prior systems.

Table 2. Isolon on two axes. Empty cells are not claimed.

	Ghost model	Executable module	Binary on target
L3	<code>ept_model</code> exclusivity	—	—
L2	—	<code>memory/ept.rs</code>	—
L1	—	Kani on unsafe	R640 Linux+virtio+SMP

Fallback is Verus $\rightarrow$ Kani $\rightarrow$ asserts and fuzz. Shipping is allowed at L1/L2 when L3 is still open, provided the paper names the cell.

Proven Core modules follow a four-file layout: executable Rust, a spec file, a proof file, and tests (Kani + unit). The exclusivity theorems of record do not use that layout. They live in one `verus!` block in `ept_model`. Putting them beside live `memory/ept.rs` would imply the EFI crate is L3. It is not.

4 Exclusive Ownership

4.1 Statement

For every valid EPT mapping from a guest-physical address to a host-physical frame, that frame is owned by exactly one guest: no other guest holds a mapping to it.

“Exclusively owned” means exactly one guest holds a mapping to that frame at any moment. The property must hold across map, unmap, EPT-violation handling, and live-migration page transfer.

Hypervisor separation—the informal sentence “belongs to neither the hypervisor nor any other guest”—is *not* this theorem. Guest id 0 is reserved; hypervisor frames are modelled as absent from *owned*. Absence is definitional. We list it in Section 2.3 next to the SDM and the SMT stack. The ghost crate never mentions the EFI’s own page tables. A bug that installed a guest leaf pointing at `.text` would not be caught by these lemmas.

4.2 Ghost state

The ghost model is two finite maps (Figure 1):

- $owned : FrameId \rightarrow GuestId$
- $by_gpa : (GuestId \times Gpa) \rightarrow FrameId$

Guest id 0 is reserved (hypervisor / unmapped). 4K steps require page-aligned GPAs. A map step is enabled only when the frame is free and the (guest, GPA) pair is free. An unmap step is enabled only when that pair is present.

The predicate (from the crate, names only) is:

$exclusive_ownership(m)$ iff every owned frame has a `by_gpa` witness, every `by_gpa` entry has a matching owner,

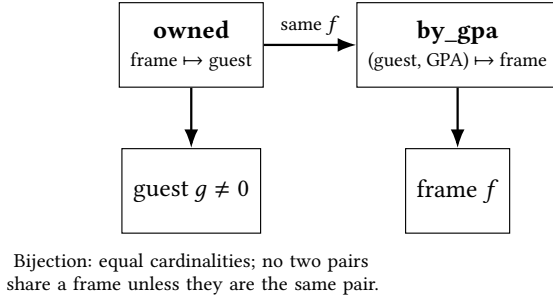


Figure 1. Ghost EPT ownership registry. Exclusive ownership is four conjuncts: every owned frame has a matching by_gpa witness; every by_gpa entry has a matching owner; a frame is the image of at most one pair; the two maps have equal length.

a frame is the image of at most one pair, and the two maps have equal length.

Empty maps satisfy it. A hypervisor frame is modelled as *not* being in *owned*: the ghost registry only tracks guest-held frames.

4.3 Steps

A 4K step is $\text{Map}\{g, a, f\}$ or $\text{Unmap}\{g, a\}$. Enabled steps preserve exclusive ownership (one-step lemma). Any finite sequence of enabled steps preserves it. Guest ids are already a field of each step, so the sequence theorem is already an N -guest statement. A two-guest lemma maps two distinct non-zero guests onto distinct free frames.

Large pages are spans of 4K frames (512 for 2M, 2^{18} for 1G). Span map/unmap lemmas lift exclusivity to 2M/1G. Hardware 2M-leaf decode lemmas relate a leaf PTE encoding to an identity-map intent. A ghost SRAT/SLIT records which node a mapped frame sits on; SLIT distance never appears in the four exclusivity conjuncts, so affinity is orthogonal bookkeeping, not part of the isolation theorem. None of these lemmas are a full multi-level EPT walk of the EFI.

4.4 EPT-violation dispositions

An EPT miss is modelled as one of three dispositions:

- **EmulateNoMap** – emulate MMIO (APIC / virtio) without installing a mapping. Registry unchanged.
- **Reject** – fail closed. Registry unchanged.
- **ClaimMap** $\{g, a, f\}$ – demand-fill a 4K map, enabled only when that map step would be enabled.

Every enabled disposition preserves exclusive ownership. Emulate-then-claim is a derived post.

Table 3. Named exclusivity theorems in the host-only crate.

Theorem	Claim
single_guest_4k_map_unmap	finite 4K sequences
n_guest_4k_map_unmap	CI alias of the previous row
concrete_*.guest_refine	scoped ghost $\leftrightarrow$ exec
large_page_map_unmap	2M/1G spans
numa_map_unmap_affinity	node affinity (orthogonal)
hw_2m_leaf_refines_identity	2M leaf encode
ept_violation_preserves	miss dispositions
page_transfer_preserves	unmap $\rightarrow$ map handoff
alloc_map_unmap_refines	pool + EPT refine

4.5 Page transfer

Live migration of a frame is unmap-at-source then map-the-same-HPA at destination. The step is enabled when source holds the GPA, destination GPA is free, and guests are distinct and non-zero. Posts: destination owns the frame at the new GPA; the source mapping is gone; exclusivity holds. This is a ghost primitive. It is not vCenter vMotion, not a multi-host protocol, and not a proof that the product migrate engine is correct. That engine is outside the Proven Core.

4.6 Theorems of record

Table 3 lists the named theorems. Supporting lemmas (empty-exclusive, map-already-owned-unchanged, span preservation, mock bring-up facts) are in the same crate. The 80/0 CI line is in Table 4.

4.7 Proof sketch: map preserves exclusivity

We sketch the 4K map case; unmap is the inverse bookkeeping. Assume *exclusive_ownership*(m), $g \neq 0$, a 4K-aligned, $f \notin \text{dom}(\text{owned})$, and $(g, a) \notin \text{dom}(\text{by_gpa})$. Let $m' = m.\text{ghost_map}(g, a, f)$.

Witness. For the new frame f , the pair (g, a) is the witness required by conjunct 1. Old frames keep their old witnesses because we did not delete by_gpa entries.

Owner match. The new by_gpa entry points at f , and $\text{owned}'[f] = g$. Old pairs are unchanged.

Uniqueness. Suppose two pairs in m' share a frame. If that frame is f , both pairs must be (g, a) because f was free. If not, uniqueness in m applies.

Cardinalities. Both maps gain one key, so lengths stay equal.

Verus discharges this with map lemmas from `vstd` (broadcast `group_map_lemmas` / `group_set_lemmas`). The one-step lemma case-splits on Map vs Unmap. The sequence theorem is induction on Seq length: the empty sequence is trivial; the inductive step applies the one-step

lemma and recurses. The N -guest CI name is that same induction.

4.8 What the obvious formulation gets wrong

Violation dispositions. The miss handler is where every hypervisor decides whether a GPA is RAM or a device. Demand-fill—every EPT violation becomes ClaimMap—is the natural first implementation: the guest touched it, so map it. On Isolon that would install a leaf for APIC and virtio MMIO GPAs, either aliasing device memory into the ownership registry or handing the guest a host frame for a region that should stay unmapped. A later guest that maps the same HPA would then see AlreadyOwned for a frame that should never have been guest RAM. The trichotomy (EmulateNoMap / Reject / ClaimMap) is the systems fix, independent of Verus: classify the GPA before installing a leaf. MMIO is emulated with the registry unchanged; fail-closed is identity on the maps; demand-fill is enabled only when the 4K map step would be. Emulate-then-claim is a derived post. Any EPT/NPT miss path has the same fork. The bug class is missing the fork, not a missing SMT lemma.

2M/1G spans. The obvious encoding of a 2M map is a loop of 512 4K maps. That is the wrong object. SMT would need a loop invariant over 512 iterations, still would not know that those 512 keys are *one* leaf with the page-size bit, and would still be 512 ghost steps rather than one hardware PDE. What we wrote instead is a span predicate (every aligned 4K key in the region is free) followed by a bulk insert, plus an arithmetic lemma that 512 aligned 4K frames constitute a 2M leaf (2^{18} for 1G). Those facts are still not a hardware page-walk. They are why “map 4K k times” is not the 2M theorem.

SMT as bug finder. We did not catch an executable aliasing bug with Verus or Kani. The ghost model was written against a registry that already rejected double-map of the same HPA. Kani’s bounded no-double-map and allocator no-alias harnesses pass. Runtime on iron found real bugs—Enable-XSAVES, INVPCID, WaitForEvent state pokes (Table 8)—none of which are exclusivity lemmas.

Violation ClaimMap reuses the 4K map lemma. Emulate and reject are identity on the registry; the proof is $m' = m$. Transfer is unmap then map of the *same* frame: after unmap, f is free, so the destination map is enabled. Posts are the destination owner and the absent source key.

5 The Ghost / Executable Gap

A discharged ghost model is not a proved .efi.

Live EPT (memory/ept.rs, 705 lines) maintains an ownership registry and asserts exclusive-ownership-shaped facts

at map/unmap. Beside it sit 357 lines of spec/proof notes. Those notes are not the 80-function crate. The host-only crate exists *because* linking Verus into the UEFI target is the wrong packaging: proofs run on a Linux CI host under a pinned Verus release; the EFI is a UEFI target, not a Linux sysroot.

Scoped refinement lemmas show that a concrete map representation, under a restricted interpretation, commutes with ghost map/unmap for 4K (one guest, then N guests) and that an allocate-then-map path refines. They are not a bisimulation of VMXON, VMCS host state, EPT pointer load, and every MMIO emulator the EFI contains.

Consequences:

- L3 is the ghost crate. Live EPT is L2.
- VMX, VMCS, hypercalls, MSR firewalls, audit integrity, and IPI confinement are in the budget in Table 1 and are not at L3 beside ept_model1.
- Iron serial logs raise confidence in the *runtime path*. They do not raise Verus maturity. Lab L3 and R640 L1 are both real; they are not the same claim.

5.1 Live registry versus ghost maps

The live engine is not a Verus Map. It is a bounded table: EptMap holds at most 512 explicit 4K claims in production builds (8 under Kani, so CBMC can unwind), plus a 16-slot range registry for the precise identity window used at Linux bring-up.

Collision and exhaustion are different errors. Map or range-claim of a taken frame or overlapping span returns AlreadyOwned. When every slot is occupied, both tables return Full. There is no silent drop and no fallback from 4K slots into range claims.

Linux bring-up does not fill the 512-slot 4K table. Guest e820 is 256 MiB; the precise identity window is $[0, \text{PRECISE_BYTES})$ with $\text{PRECISE_BYTES}=512$ MiB (131,072 4K pages). The window matches QEMU -m 512M: it covers guest page tables, the early frame pool, and 256 MiB of guest RAM, while leaving the local APIC at $0xFEE00000$ unmapped by omission. The upper 256 MiB is the BAR and shell window: virtio BARs and G1–G3 2M holes sit above e820 and inside the 512 MiB identity map. Hardware identity uses 2M leaves (256 leaves).

The executable counterpart of that window is the range registry, not the proved 4K table. With no extra-guest holes, bring-up inserts **one** range covering the full 512 MiB for the Linux guest (host test: $\text{EptRangeMap} : \text{len}() = 1$). Multi-VM probes punch private HPA holes above e820 and claim those spans for shell guests. The four-VM case is five range rows: G0 below e820, three 2M shells, G0 remainder (host test: $\text{len}()=5$). Pairwise HPA non-overlap

on that path is `claim_range` returning `AlreadyOwned` on overlapping spans, plus a sorted-hole cursor that rejects `hpa < cursor` as `Invariant`. After insert, the boot path checks that G0 owns GPA 0 and the last e820 page, and that each hole's guest owns its base while G0 does not. The host test additionally checks that G1 does not own G2's base, G2 does not own G3's, and G3 does not own G1's. It does not enumerate every guest-pair of the 512-slot 4K table. The 4K self-test is a *different* object: two-guest alias on `EptMap`. Iron exclusivity among the four guests is therefore hole arithmetic in the 16-slot table, L1-tested, not a ghost lemma.

Scoped refinement lemmas interpret the 4K table into ghost maps. They do not interpret the range registry. Production guest-physical coverage at Linux boot—the 512 MiB identity window, including the 256 MiB e820—goes through the range path. That path has no correspondence in the proved model. This is not a gap between two encodings of the same mapping mechanism; it is disjointness. The 4K table is the demand-fill and self-test path (map a bring-up guest, then assert a second guest cannot alias the same frame), including MMIO-not-mapped misses that take `ClaimMap`. If a 513th distinct 4K claim is required, the live table returns `Full` while the ghost theorem still holds of an abstract `ghost_map`. That is why Table 2 leaves the executable-module L3 cell empty.

A range claim is a span. The crate already has span predicates for 2M/1G. Modelling a range row as a ghost step, with pairwise HPA non-overlap among the 16 slots and against 4K rows, is the lemma that would put the production path in the same artifact column as the proof. This paper does not claim that lemma.

5.2 What would close the gap

Four artifacts, none of which this paper claims:

1. A ghost range-claim step whose posts are pairwise HPA non-overlap among range rows and against 4K rows, so the Linux window is in the proved model.
2. A proved executable EPT walker whose leaves are in 1–1 correspondence with `by_gpa`, including the HV-vs-guest split (“belongs to neither” as a fact about HV page tables, not as absence from a guest registry).
3. An IOMMU / DMA story: devices we do not model cannot write a guest-owned frame or a hypervisor frame.
4. Linking or replaying (2) on the UEFI target, or a refinement from the EFI binary's EPT root pointer to the ghost maps.

Until then, L3 names the ghost crate. The EFI is not L3.

6 Evaluation

We evaluate five questions. (1) Does the ghost model verify? (2) How large is the proved surface? (3) What covers guest-physical space at Linux boot? (4) What does the runtime do on a lab host versus a real R640? (5) Does the two-axis ladder change how landmark abstracts read?

6.1 Proofs

Toolchain: Verus pin 0.2026.07.12 (asset 0b42f4c; Linux zip SHA-256 f6f4f5d0. . . fbd01c0 in-tree; CI never uses latest) and Kani 0.67.0 (CBMC backend). The exclusivity crate is verified with `cargo verus verify`. Against that pin the crate reports **80 verified, 0 errors**, including the *N*-guest alias, with no `admit` on the exclusivity path. Wall-clock and peak RSS for that run are not in Table 4. Kani covers bounded unsafe in the live engine (`MAP_CAP = 8` unwind). That is L1 on unsafe, not L3 on exclusive ownership.

6.2 Size and occupancy

Table 4 is tree- and kit-measured. The iron EFI that reached Linux shell plus the M4 probe chain is 1,085,952 bytes (1.04 MiB), against a 15 MB target and a 20 MB hard limit. A later in-tree release build is 1.62 MiB. Rust sources in the tree total 64,140 lines (`wc -l` of `*.rs`, excluding `target/`): about 40,500 product, 15,600 gate harnesses, 8,000 tests. The Proven Core *budget* remains 15,000 lines (Table 1); the surface at L3 is the 2,366-line host-only crate. Table 5 is the EPT slice.

Range occupancy is from the host test of the same `claim_precise_*` function the boot path calls. Linux-only: one range row covering 512 MiB. Four-VM: five rows (G0 below e820, three 2M shells, G0 remainder). Hardware uses 256×2M leaves. Guest e820 is 256 MiB. COM2 has no slot-counter dump. VM-exit cycles and boot-to-shell versus KVM on the R640 are empty cells.

6.3 Runtime: lab host versus R640

Table 6 is the split. Cells are contemporaneous serial or host-test evidence, not a wish list. Nested VT-x is a third column because operators (and reviewers) otherwise collapse it into iron.

On that R640 (iDRAC virtual media, COM2): the EFI archives VMXON, a 512 MiB identity window, unmodified Linux 6.12 through `/init` to a shell, then—on that same boot—four-VM probes (Linux G0 plus three SHELL guests), `virtio-blk`, `virtio-net`, dual-vCPU SMP, and VMX-OFF. Secondary Enable-XSAVES was required on Xeon Scalable; without it the guest init hit a stack guard. Firmware SNP/Tcp4 after `ExitBootServices` is a platform limit of the

Table 4. Measured quantities. Empty cells are unmeasured.

Quantity	Value
Iron .efi (Linux+M4 boot)	1,085,952 B (1.04 MiB)
In-tree release .efi	1.62 MiB
Size target / hard limit	15 MB / 20 MB
Tree Rust LOC (*.rs)	64,140
Proven Core budget / L3 crate	15,000 / 2,366
Verus pin zip SHA-256	f6f4f5d0. . . fbd01c0
Verus functions (incl. alias)	80 verified, 0 errors
Linux-only range occupancy	1 row (host test)
Four-VM range occupancy	5 rows (host test)
Hardware 2M leaves in the window	256 (131,072 4K pages)
Guest e820 RAM / identity window	256 MiB / 512 MiB
4K table cap / range slots	512 / 16; exhaustion = Full
VM-exit cycles vs KVM on R640	—
Boot-to-shell wall-clock vs KVM	—
verus verify wall-clock / RSS	—

Table 5. Line counts (wc -l) for the EPT-related surface.

Path	Lines	Role
ept_model crate	2,366	L3 ghost + proofs
memory/ept.rs	705	live EPT (L2)
ept_spec + ept_proof	357	notes beside live
frame allocator + spec	362	allocator + ghost set

virtual-floppy path; durable HTTP is a host-owned Broadcom LOM. The SPA can VMLAUNCH a private-EPT SHELL guest and re-enter; that guest is a CPUID stub, not a distro. Guest firmware issued ATAPI READ(10) and booted El Torito CD EFI. Persist-detect stamps on a USB stick are LBAs, not a guest root filesystem.

On nested VT-x we authored guest firmware after third-party WaitForEvent forcing faulted. That firmware completed an alpine-extended setup-disk -m sys (installer printed “Please reboot”) and reboot-to-disk. Nested virtualization is not the R640. The USB stick used on iron is too small for that distro image. Host CI and nested runs do not print an install-complete marker.

6.4 Operational gates

72-hour soak metrics on a laptop, PDF audit-report templates, and a bring-up spec review closed on the lab host. They are L0/L1 process, not L3. Iron distro install is a product gate, not a requirement of the ghost theorem.

6.5 Labelling the literature

Table 7 applies the two-axis grid to seven prior systems plus Isolon. Cells are a reading of each paper’s abstract and stated artifact, not a re-proof.

Landmark abstracts typically name an executable artifact: seL4’s C kernel [14], then the ARM binary via translation validation [23]; CertiKOS’s C implementation [8]; Komodo’s monitor [6]; Hyperkernel’s simplified kernel [22]. XMHF and Verisoft XT name fragments of an executable hypervisor, not a whole binary [18, 27]. SeKVM’s abstract presents “a formally verified Linux KVM hypervisor”; the Coq development is an overlay on commodity KVM [19, 20]. That is the closest analogue to Isolon-shaped slippage: an L3 claim that a reader can place on the wrong artifact column. Even there the proved object is executable code, not a ghost crate.

The finding is not that the literature’s labels are systematically false. It is that a one-axis “L3 isolation” label would put Isolon’s crate in the same cell as seL4’s C kernel, which is the confusion the instrument exists to prevent. Isolon is the only row whose L3 result and whose shipped mapping path occupy different artifact columns.

6.6 Artifact

We commit to a supplementary zip: the ept_model crate, the pinned Verus zip SHA-256, a README that reproduces 80/0, and a container or exact install command for that pin. Anonymized COM2 excerpts for every row of Table 6’s R640 column (product markers stripped) belong in the same zip. The paper stands alone if the reviewer ignores the zip.

7 Experience: Five Lessons that Generalized

The following are the lessons we would tell a group starting a verified Type-1 hypervisor in Rust [5]. They are the spine of the paper. The ghost model is an artifact inside this story.

7.1 Pin the prover or the proof is a diary

Verus and its SMT backend move. We froze the tagged Verus release 0.2026.07.12 (asset 0b42f4c) with a SHA-256 of the Linux zip in-tree, and forbade CI from fetching latest. The 80/0 line is replayable only against that pin. Groups that verify “on nightly” will not be able to say, a quarter later, whether a new admit is a regression or a toolchain change. Budget pin maintenance as a first-class task, not as hope.

7.2 The UEFI target cannot run SMT

Isolon’s EFI target is UEFI PE/COFF, not a Linux sysroot. Verus wants a host sysroot. We stopped trying to link proofs into the PE/COFF image. The 2,366-line crate lives on the CI host; the 705-line engine lives in the EFI. That split is why

Table 6. Split evaluation. Nested VT-x $\neq$ PowerEdge. Host Verus $\neq$ EFI. The Lab/CI column is where the prover runs; the R640 column is the only iron.

Surface	Lab / CI	Nested VT-x	PowerEdge R640
Verus exclusivity (80 / 0)	yes (L3)	—	not claimed
Kani on unsafe	yes (CI pin)	—	not claimed
VMXON + VMEXIT	yes	yes	yes
EPT identity + Linux 6.12 shell	yes	—	yes
LAPIC timer re-entry / IRQ inject	yes	—	yes (same boot)
virtio-blk probe	yes	CD+virtio	yes (same boot)
virtio-net + vSwitch probe	yes	—	yes (same boot)
Dual-vCPU BSP+AP (shared EPT)	yes	—	yes (same boot)
Four VMs: Linux + three SHELL stubs	yes	—	yes (same boot)
Durable host-NIC HTTP after EBS	lab NIC	—	onboard LOM
SPA launches a SHELL guest	host smoke	—	stub, not a distro
Guest firmware CD EFI (El Torito)	—	CD+virtio	yes
Distro sysinstall + reboot-to-disk	—	yes	open
Persist-detect LBA stamps on USB	—	—	stamps, not a rootfs

Table 7. Retrospective (maturity, artifact) reading. Isolon is the row where L3 and the shipped mapping path are different cells.

System	Artifact	Mat.	What the abstract names
seL4 [14]	executable C	L3	kernel C implementation
seL4 binary [23]	ARM binary	L3	translation validation
CertiKOS [8]	executable C	L3	C contextual refinement
Komodo [6]	executable monitor	L3	monitor implementation
Hyperkernel [22]	executable kernel	L3	simplified syscall iface
SeKVM [20]	executable overlay	L3	Linux KVM (overlay in Coq)
XMHF [27]	executable fragments	mixed	core memory integrity
Verisoft XT [18]	selected C	mixed	Hyper-V components (VCC)
Isolon exclusivity	ghost model	L3	this paper
Isolon memory/ept.rs	executable module	L2	this paper
Isolon R640 boot	binary on target	L1	this paper

L3 and L2 are different columns in Table 6. A single-crate “verified hypervisor” that does not build on the target you ship will still be described as verified by someone; make the packaging match the claim first.

7.3 Do not poke third-party firmware

When guest firmware blocked in `WaitForEvent`, a tempting fix is to poke its wait object, skip a check, or inject a fault. We did that on a development path. The firmware then page-faulted on a `NULL` event pointer and entered a dead loop. We do not own that firmware’s invariants; forcing one of them violates the next. The product path is firmware we author, driven only through architected inputs (interrupts, device registers, DMA the spec says is ours). Device emulation remains outside the Proven Core, so a firmware bug is still

not an EPT exclusivity theorem—but it is no longer a self-inflicted one. This lesson is independent of Isolon: if you do not own the blob, do not rewrite its control state.

7.4 Firmware TCP is not a management plane

iDRAC virtual floppy advertised PXE/HTTP but not a `Tcp4` service binding. Firmware SNP that had accepted a listen before `ExitBootServices` did not survive it. Durable HTTP on the R640 is a host-owned Broadcom LOM after the hypervisor owns the NIC, not a firmware socket. Quantitative split: PRE-EBS HTTP closed on SNP as a bring-up window; post-EBS HTTP closed only after a native driver. Papers that claim “the hypervisor serves a Web UI” should say which NIC stack and which side of `ExitBootServices`. In-process REST on a lab executable is a third, still weaker, claim.

Table 8. Archived negative results. None of these are L3.

Failure	What it blocked
Guest /init stack guard without Enable-XSAVES	Linux shell on Xeon Scalable
Firmware Tcp4 absent on virtual floppy	“HTTP forever” via SNP
SNP listen dead after ExitBootServices	post-EBS firmware UI
WaitForEvent state poke → #PF	third-party firmware as product ISO path
USB persist LBA stamps	calling stamps a root filesystem
USB stick capacity < alpine-extended	iron sysinstall on that stick

7.5 Nested VT-x is not the production server

Nested alpine-extended sysinstall completed: installer printed “Please reboot,” then reboot-to-disk succeeded. That is a real product-path result on nested VT-x. The USB stick used on iron is too small for that distro image; El Torito CD EFI is not setup-disk. A lab hypervisor that installs Linux under nested VT-x has not demonstrated ISO install on the machine in the rack. Table 6 keeps those columns apart.

7.6 Negative results we archived

Table 8 lists failures we kept. None of them are L3.

The XSAVES miss is the most transferable: a VMCS secondary-control bit that is optional on a laptop CPU and required on the server. Isolation proofs do not mention it. The guest still does not boot. Treat CPU-feature bring-up as L1 evidence, not as a corollary of exclusive ownership.

8 Related Work

Verified systems and their unverified edges. Fonseca et al. [7] audited IronFleet, Verdi, and Chapar from the outside and found 16 bugs, not in protocol logic but at the interface of verified and unverified components—specifications, shims, and tools. That is this paper’s thesis, at this venue, from the other direction: Isolon reports the same class of gap from inside one Type-1 hypervisor and proposes a (maturity, artifact) label so an abstract cannot place an L3 ghost model on the binary that left the compiler. Table 7 is the evaluation of that instrument.

Proved kernels and enclaves. seL4 [13, 14, 21] is the landmark machine-checked microkernel. The seL4 team was careful about proof assumptions—hardware, compiler, assembly, boot—and later closed the C-to-binary gap on ARM by translation validation [23]. Isolon borrows the discipline and a fraction of the scope: we do not prove the scheduler or

the device model, and we do not have a binary-level theorem. CertiKOS [3, 8] builds a certified concurrent kernel with contextual refinement. Komodo [6] verifies a software enclave monitor on ARM TrustZone. Ironclad Apps [9] push verification out to application principals. These systems show isolation *can* be proved. They do not ship as B00TX64. EFI on iDRAC virtual media.

Hypervisors with a verification story. Hyperkernel [22] pushes a hypervisor toward push-button verification with a simplified interface. XMHF [27] is an extensible hypervisor framework with selected proofs. Verisoft XT applied VCC to Hyper-V components [18]. SeKVM / MicroV work [19, 20] layers formal memory protection onto a commodity KVM. Isolon is the opposite packaging: a new UEFI binary with a tiny proved ghost registry, not a verified overlay on Linux.

Small-TCB virtualization. NOVA [25] and related microhypervisor designs argue for a small TCB in the EuroSys tradition. Isolon’s cut is the Proven Core list in Table 1. OKL4-style microvisors [10] similarly shrink the host. Production hypervisors—ESXi [2], Xen [1], KVM [12]—isolate with EPT/NPT and years of testing.

Tooling. Verus [16, 17] is the SMT prover we pin. Kani [4, 15] covers unsafe; it compiles MIR to CBMC. VanHattum et al. [26] is the ICSE-SEIP paper on dynamic trait objects in that pipeline, not a substitute for the tool or for CBMC. Push-button verification of file systems [24] is the methodological cousin of Hyperkernel.

9 Limitations

- Live executable EPT is L2. L3 is the host-only ghost model. Ghost↔exec refine is scoped, not a full EFI bisimulation. The range registry that covers Linux GPA has no ghost correspondence.
- Most Proven Core modules in Table 1 are not at L3.
- Large-page and NUMA L3 are lab/CI. Iron stopped at multi-VM probes for those surfaces.
- PageTransferStep is a ghost primitive, not live migration product.
- DMA confinement and IOMMU are future work. Exclusive ownership is conditional on no passthrough and emulation-mediated DMA.
- Nested distro sysinstall is complete and is not iron.
- Iron installer, TLS, and guest console remain product residuals.
- Verus upgrades can break proofs; pin maintenance forbids latest.
- SMT found no executable aliasing bug.

- No KVM cycle comparison on the R640; no *verus* verify wall-clock in this PDF.

Threats to validity. Iron evidence is one PowerEdge R640 and iDRAC virtual media, not a fleet. Nested VT-x is one lab host. The 80/0 line is one pin of one prover; we have not replayed under a second SMT solver. LOC counts are `wc -l`, not a proven TCB. Range occupancy is a host test of the boot claim function; COM2 has no slot-counter dump. Table 7 is our reading of published abstracts, not a re-verification of those systems. Experience lessons (XSAVES, SNP, firmware pokes) are one implementation’s failures; they follow from Intel VMX, UEFI ExitBootServices, and “do not own that blob,” not from Isolon-specific APIs.

10 Conclusion

Isolon is a UEFI Type-1 hypervisor that boots unmodified Linux 6.12 with virtio and SMP on a PowerEdge R640, and an experience report about the gap between that binary and a machine-checked isolation claim. The two-axis ladder (maturity $\times$ artifact) is the instrument; Table 7 applies it to Isolon and to seven prior systems. Isolon’s L3 cell is a guest-exclusivity ghost model. Live EPT is L2. The R640 boot is L1. The 512 MiB Linux window is a range registry with no ghost correspondence. Virtio rings are guest frames the emulator touches; the property is about mappings. Nested authored firmware completed a distro sysinstall. Nested is not iron.

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